

The Influence of Sponsorship on Open-Source Software Developers' Activities on GitHub

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Abstract—Studies on the OSS communities have shown that financial supports are critical to OSS developers and projects to maintain their progress and sustainability. However, there were few developers being paid directly for maintaining OSS projects in the past. The GitHub Sponsors program that brings financial supports to the general OSS developers in GitHub—the world's largest OSS platform may make a difference on this situation in the future. In this paper, we present a data set on GitHub Sponsors and conduct a data-driven study to analyze the participants of the program and the impact of sponsorships to developers' activities and their projects' outcomes and qualities. The results of our survey suggest that most developers state they will contribute more with sponsorships and provide some privilege for their sponsors. And through quantitative study, we find that developers make more contributions on GitHub after they got/offered sponsorships. Moreover, gaining sponsorship also has a weakly positive impact on developers' collaborators that did not get sponsorship. And not only developers, but their own or contributed projects also can be motivate by sponsorships. Our findings are useful to the community by understanding the impact of sponsorships on users' activities and projects' progress and sustainability, and helping the managers to improve the current financial support mechanism.

Index Terms—GitHub Sponsors, Open Source Software, Financial Support, Data-Driven Study

I. INTRODUCTION

The importance and impact of providing financial support to open source software (OSS) developers and projects have attracted much research interest [1]–[5]. While many studies [3], [4], [6], [7] have identified that intrinsic motivation—self-determination and enjoyment—is the most important in driving OSS contributors, there are evidences showing that many of the OSS contributors also wish to get paid by maintaining OSS projects. For example, according to studies conducted in [1], [2], OSS contributors wish to ‘*earn a reasonable livelihood from their efforts*’. According to the interview provided by GitHub [8], OSS maintainers hope to ‘*making maintaining open source software a full-time job*’. However, OSS communities hardly receive direct payments from contributing in OSS [3], [6]. Among the 4300 participants of the survey conducted by DigitalOcean, only 25% report their company donates more than one thousand dollars every year to OSS projects [9], which does not necessarily turn into payments to the developers.

On May 23rd, 2019, GitHub announced the beta version of GitHub Sponsors program [8] that allows its users to provide

monthly financial supports to the developers on GitHub. Unlike one-time donations studied by most existing works [10], the new mechanism introduced by GitHub Sponsors allows the sponsors to provide regular support every month and cancel the sponsorship anytime. This creates an opportunity to observe the effect of providing constant financial support to the activities to the developers, and also to the contributions of projects. So we conduct a data-driven empirical study on developers who participate in the GitHub Sponsors and projects related to them to demonstrate the behavioral affects of sponsorships on open source software developers and projects. Specifically, we review the sponsored developers' “sponsor pages” as a survey to do the qualitative study and conduct statistical tests for quantitative analysis. And this work aims at answering the following research questions.

RQ1: Who are involved in the GitHub Sponsors program? We analyze the profiles of *sponsees* (GitHub users who are being sponsored) and *sponsors* (GitHub users who provide sponsorships), and discover that the *sponsees* are often senior members of GitHub who are active contributors with many watchers, and the *sponsors* are often younger and have less contributions than *sponsees*. Some *sponsees* and *sponsors* register after the program is announced. There is a global network of sponsorships forming in GitHub, while most of them exist in western developed countries.

RQ2: What are the impacts of sponsorships on developers' contributing activities on GitHub? We review developers' “sponsor page” and find that most *sponsees* state they will devote more time to OSS community and provide some privileges for *sponsors*. And we conduct quantitative studies to compare the level of contributing activities of the *sponsees*, *sponsors*, and *collaborators* (GitHub users in the same organizations with the *sponsees*) before and after establishing sponsorship relations. The results suggest that sponsorships do promote developers' contributions in public repositories: *sponsees* are more concentrated on their own work and become more earnest; *sponsors* are also more active and it seems that they get some priorities of issues submitted to their *sponsees*' projects; and the collaborators show the same change in contributing activities with the *sponsees*.

RQ3: What are the impacts of sponsorships on the sponsees' projects in terms of outcome and quality? We mine the OSS projects owned or contributed by the *sponsees*, and compare the projects' outcome and quality before and after

the *sponsees* have received sponsorships. We find that after the *sponsees* have started receiving sponsorships, the projects they owned or involved in become more productive and healthier with less bugs reported. The results suggest that sponsorships to developers have a positive impact on their projects.

Our study also has some implications. Reviewing sponsored developers' "sponsor pages" can be an easy way to conduct qualitative analysis, and we can get categories from the data instead of using preconceived categories that do not apply well on this topic. And through our quantitative study, those users who want to get priorities from developers can give better consideration on whether they should provide sponsorships. In summary, this work makes the following contribution: we conduct a data-driven empirical study and get a series of findings, which provide insight into how consistent financial stimulations impact developer behavior and their related projects, especially in positive ways.

II. BACKGROUND & RELATED WORK

In this section, we provide the background and summarize works closely related to this work.

A. GitHub Sponsors

The GitHub Sponsors program was announced on May 23rd, 2019 [8]. Later on Nov. 4th, 2019, the program was out of beta and operating in 30 countries and regions in the world [11]. With this program, developers on GitHub can receive payments directly from other users for their contributions in OSS projects. For sponsored developers accepted by the program, GitHub doubles their sponsorships with the GitHub Sponsors Matching Fund [12] that offers up to \$5000 USD in their first year of receiving sponsorships. After half a year of operation, the program enabled organization accounts to receive funding in 34 regions on May 12th [13].

To become a sponsored developer on GitHub, one need to join the "waitlist" by submitting an application which will be reviewed by GitHub. The user who has a bank account in a country or region where GitHub is available will receive an email from GitHub when the application is reviewed. Once the application is accepted, there will be a "Sponsor" button on the user's profile page, by clicking which will show the user's "sponsors page" that includes the user's self-introduction (Fig. 1(a)), and tiers of sponsorship (Fig. 1(b)). In addition, a list of the developer's public sponsors and pinned projects are also shown on the page which are omitted in Fig. 1. Different from crowdfunding platforms like opencollective.com, rewards for sponsors are not mandatory. However, the developers often offer some privileges to their sponsors as described in the tiers of sponsorship (Fig. 1(b)).

The *sponsors* are other users on GitHub. Everyone wants to give financial support or express appreciation can be a *sponsor*, as long as he/she has an available account. Choosing a tie of sponsorship at sponsored developers' "sponsors page", you can be a *sponsor* of him/her. And the payments will be deducted automatically on a fixed day of every month. If the *sponsor* wants to stop sponsoring, he/she can canceled the

URL: <https://www.github.com/sponsors/<username>>

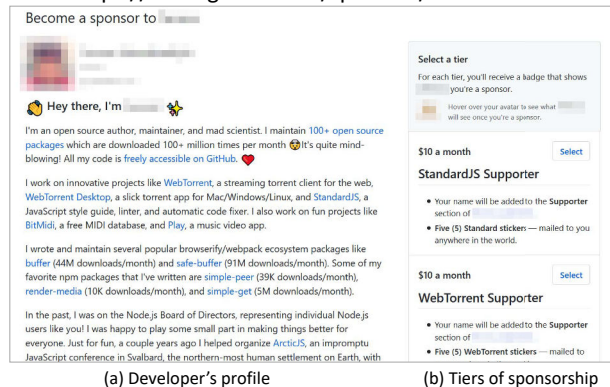


Fig. 1. The 'sponsor page' for a sponsored developer on GitHub.

sponsorship at his/her own administrator page at anytime. Different from one-time donations studied in existing works, the sponsorship relationship lasts after becoming a *sponsor*, and breaks after the *sponsor* stops providing monthly sponsorships.

B. Related Work

This work is closely related to works about the motivations of OSS developers and the effect of providing financial supports. The motivations of OSS developers have attracted much research interest [3], [4], [14]. Lakhani et al. [3] discover that enjoyment-based intrinsic motivation, instead of extrinsic benefits, is the strongest and most pervasive driver of effort in OSS projects. In [4], Roberts et al. study the intrinsic and extrinsic motivations of OSS developers and model the relations between the developers' motivations and performance. They discover that being paid to contribute is positively associated with the level of contributions to the source code [4]. Despite that few developers are paid directly for maintaining OSS projects [3], many contributors wish to receive financial support for their efforts [1], [2].

With respect to monetary support to OSS developers, existing studies [2], [15]–[19] explore payments and donations made to the developers, and analyze the impacts of financial rewards to OSS projects. Atiq et al. [17] point out that the importance of financial support for the progress of OSS projects is well-recognized by the research community. And they find that proper management is essential to the projects' sustainability, even if the financial benefits received by the developers are unequal. In [2], Krishnamurthy et al. adopt the private-collective innovation model to describe the acceptance of monetary rewards for OSS developers. They highlight the importance of studying the acceptance behavior for OSS related researches, and relate the behavior to various motivational components. Nakasai et al. analyze the donations made to the Eclipse project [16]. They argue that making donations is also an important contribution besides contributing to the source codes. And their analyses suggest that new releases attract more donations but bugs negatively affect the amount of donations. Similar to the above work, Zhauniarovich et al.

[15] conduct a quantitative study exploring the relationship between the amount of donations received by a repository and its hosting-related information, and find that the amount of donations is mostly associated with the number of forks. Riehle et al. [20] find that balancing the amount of paid developers with volunteer developers helps OSS projects steering. In [21], Claes et al. use random forests to predict paid and unpaid OSS developers and show the effectiveness of using machine learning techniques to identify the roles of users. As for the new financial support mechanism of GitHub, Zhang et al. conduct an empirical study on developers who participated in the GitHub Sponsors program [22]. They find that most *sponsors* provide sponsorship because of usage, and the impact of sponsorship on developers' activity is not sustainable. Moreover, they reveal some shortcomings of this program and propose a series of advice on improving the mechanism. In this work, we conduct quantitative studies on the impacts of the GitHub Sponsors program on the participants' contributing activities and projects.

III. DATA SET

In this section we describe the collection, filtering and processing of the data for our survey and quantitative analysis.

A. Data Collection

With GHTorrent [23], we can get almost all the public data we need about developers, organizations and repositories on GitHub. We also use GitHub API¹ to collect monthly sponsorship snapshots from Sep. 2019 to Aug. 2020. And we chose the snapshot on Feb. 28th, 2020 to determine the lists of *sponsees* and *sponsors*, thus giving us enough time to see the changes of their activities. Using the above lists, we next accessed users' organizations and repositories information and activity data through GHTorrent.

1) *Sponsorship snapshots*: We first retrieved users list from GHTorrent, and use the field '*sponsorshipsAsMaintainer*' of GraphQL API to select those who are sponsored developers and implement their sponsorships. We have been keeping looping through the users list since Sep, 2019 to update the sponsored users list. And by crawling sponsorships of sponsored users, we got monthly sponsorship snapshots and the sponsors list. As it takes about 1 week to traverse users list once, we do our best to make the sponsored users list complete with a delay of 1 week. In this work, we use the snapshot collected on Feb. 28th, 2020 to conduct our study.

2) *Developers and Projects*: From the snapshot taken on Feb. 28th, 2020, we find there are a total number of 875 *sponsees* and 4447 *sponsors* with 6203 pairs of sponsorship relations. As the activity data used comes from GHTorrent, so we need to exclude users who have no record in GHTorrent. After this selection, there remained 875 *sponsees* and 4245 *sponsors* with 5978 pairs of sponsorship relations. Note that through our manual check, we find there are 6 *sponsees* have changed their login on GitHub, so we merged all data of

the old account and the new one of them. Then we selected those who are in the same organizations with *sponsees* as *collaborators* from GHTorrent. As a result, there are 48041 *collaborators*. And we retrieved the list of projects owned by *sponsees* and those contributed (especially authored commits and issues) by *sponsees* at least once during our time window. The total numbers are 116080 and 139738, respectively. As there are developers who have multiple types (such as be the *sponsee* and *sponsor* in the meanwhile), and there is overlap lies between projects owned by *sponsees* and contributed by *sponsees*, so the unique user list and project list contains 51946 users and 206051 projects respectively.

3) *Sponsor Pages*: We first download 875 *sponsees*' "sponsor pages" (shown in Fig. 1), and extract the contents of self-introduction and sponsorship tiers. Then we write two copies of documents which comprised by the above contents of each *sponsees* and 5 codes described in Sec. V-B1. We rule out four developers because three of them cannot be found on GitHub, and one developer claims to have stopped receiving sponsorships while still staying in the program. As a result, a total number of 871 developers' sponsor pages are reviewed.

4) *Activity Data*: To answer our research questions, we need to collect basic information and activity data of developers and projects. First we retrieved *sponsees*' and *sponsors*' basic information to answer **RQ1**, including the creation time of account, the public Geolocation (especially country code), the number of repositories, the number of watchers of repositories, and the language of them. Then we accessed developers' and projects' monthly contributions of commits, authored and closed issues/pull-requests in recent 3 years (from Feb. 1st, 2018 to Jan. 31st, 2021) on GHTorrent. But there is a hole in GHTorrent database during June. 1st, 2019 to Dec. 31st, 2019, so we exclude data in this period. Finally, there are over 50 million commits and over 10 million issues/PRs.

B. Data Cleaning

The following steps are performed to eliminate noises in the data we used.

1) *For Developers*: We filtered out users without records and those have been deleted in GHTorrent, and exclude organization accounts which are not in the scope of our study. Then we removed users who are not active (did not have a commit in any repository) in recent 3 years [24]. For *sponsees*, we remove those whose *sponsor* has no record in GHTorrent. After filtering, there are 856 *sponsees*, 3816 *sponsors* and 40318 collaborators in our data set.

2) *For Projects*: Not all repositories in our data set are original projects, and some are unmaintained projects or projects with little issues or pull-requests. So we filtered out projects using the following rules [25]: (1) Projects must have not been deleted; (2) Projects must not be unmaintained projects [26], that is, there are at least one updated activity of the project in recent year (from Feb. 1st, 2020 to Jan. 31st, 2021); and (3) Projects must have at least one commit and issue/pull-request in our observational period (from Feb. 1st, 2018 to Jan. 31st,

¹<https://developer.github.com/v4/> and <https://developer.github.com/v3/>

TABLE I
PROFILING THE USERS.

Feature	Description
<i>AGE</i>	Age in GitHub (days) by <i>T</i> .
<i>GEO</i>	The location (country code).
<i>#REPO</i>	The number of repositories owned by the developer before <i>T</i> .
<i>#WTS</i>	The total number of watchers of repositories owned by the developer before <i>T</i> .
<i>#WTS_{AVG}</i>	The average number of watchers of each repositories owned by the developer before <i>T</i> .
<i>LNG</i>	The primary language of the developer.
# denotes ‘The number of’, and <i>T</i> denotes May. 23rd, 2019	

2021). After the filtering, there are 47183 projects remained in our data set.

C. Data Processing

We generate the monthly activities of developers and projects to conduct quantitative studies in the next section. For the activities of developers, 6 kinds of contributions are included: the commits, authored issues, closed issues, closed pull requests, merged pull requests and first delay of closing issues. And for the activities of projects, we add more 2 kinds of contributions: reported bug issues and closed bug issues. However, if a repository is forked from the other repository, it will brings duplicated commits, so only commits with unique SHA values will be kept in our data set.

Finally, there are 43803 users and 47183 projects with over 21 million commits and 9 million issues/PRs in our data set. And we computed the monthly contributions count of different types assigned to the corresponding user ID respectively for the next section.

IV. THE PROFILE OF PARTICIPATORS

First of all, we ask **RQ1: Who are involved in the Sponsors program?** We try to draw a profile of developers who took part in GitHub Sponsors program to answer this question.

A. Research Methods

We use a series of developers’ basic information to make up the profile of them, and the metrics are summarized in Table I [27], [28]. All the features in Table I are obtained in GHTorrent data set. We choose May 23rd, 2019 as the end time for all features to reflect the developers’ profiles at a fixed time point before any user can be sponsored or provide sponsorships.

There are both numerical and nominal features in Table I. For numerical features, we analyze the differences between *sponsees* and *sponsors* using box diagram and statistical measures. And for nominal features, we draw statistical charts to show the distributions of developers.

B. Results

For the numerical features, we first draw the box diagram to show the distributions of them. And the results are shown in Fig. 2. We can see that the *sponsees* are always older than

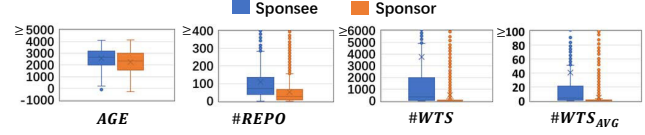


Fig. 2. The distribution of numerical features in RQ1.



Fig. 3. The distribution of different types of developers’ nominal features.²

sponsors. The minimal ages of both *sponsees* and *sponsors* are negative numbers, suggesting that some participants of the program are new users who registered after the program is announced. And the *sponsees* own more repositories and more watchers than *sponsors*. In the rest metrics, The *sponsee* also show higher value than *sponsors*.

Then for nominal features like *GEO* and *LNG*, we draw bar charts like Fig. 3 (a) and (b). We can see that the top 9 countries of both *sponsee* and *sponsor* are the same, and all of them excluding Japan are western economically developed countries. The languages used are also mostly coincident, and the top three mostly used languages are: JavaScript, PHP, and Python/Ruby. So we can infer that most developers participating in this program are geographically closed and use similar programming languages.

In summary, we find that the *sponsees* are mostly composed of senior members of GitHub and are active contributors with many watchers. And there are a few users registering because of GitHub Sponsors. Although compared with the total number of GitHub users, the number of participants is small, there is still a global network of sponsorships formed.

V. IMPACT ON INDIVIDUALS’ CONTRIBUTIONS

After profiling developers, we then ask **RQ2: What are the impacts of sponsorships on developers’ contributions on GitHub?**

It is well known that public projects and contributions propel FLOSS community forward, so it is important to study whether the sponsorships can influence developers’ public contributions on GitHub. We also want to know whether the sponsorship will motivate sponsors and collaborators of *sponsees*, so we study RQ2 which concerns the influence of sponsorship on developers’ own contributions in this section.

A. Research Methods

We first conduct a content analysis on *sponsees*’ “sponsor page” to figure out how they think sponsorship will have impacts on their contributions subjectively. Then we measure the developers’ contributing activities using a series of metrics

²country names can be found in <https://countrycode.org/>

TABLE II
METRICS TO MEASURE A DEVELOPER'S CONTRIBUTING ACTIVITIES
DURING A TIME WINDOW $\mathcal{L}$.

Metric	Description
<i>CMA</i>	The number of commits authored by the developer.
<i>ISA</i>	The number of issues authored by the developer.
<i>ISC</i>	The number of issues closed by the developer.
<i>PRC</i>	The number of pull-requests closed by the developer.
<i>PRM</i>	The number of pull-requests merged by the developer.
<i>ISCD</i>	The time delay (hours) of first closing an issue by the developer.

obtained from existing literature [28]–[30] and the metrics are listed in Table II as *CMA* to *ISCD*. And we conduct statistical analysis on three types of developers respectively.

1) *Reviewing the Sponsor Pages*: After the analysis of Sec. IV-A, we have a preliminary understanding on developers' characteristics. And to know what the attitude of *sponsees* towards sponsorships, we review their "sponsor pages". We follow the steps of content analysis [31] to perform the study as follows. First, two researchers (i.e., the first and second authors of this paper) read their sponsor pages independently to find clues on how they will change in behaviors or provide rewards to the sponsors.

Second, a coding frame is devised to describe the potential impact of sponsorship on the developers' activities by summarizing the two researchers' observations. We focus on the impact on the developers' contributing activities and omit cases of sending gifts or providing opportunities to gain more publicity for the sponsors (e.g., to include the sponsors' names in README files). We also focus on responses targeting at the sponsors in addition to general descriptions such as "[The developer] will spend more time on open source". The resulting code book is described in Sec. V-B1.

Third, the two researchers independently assign the codes to all of the *sponsees* by reading their sponsor pages. The Kappa test is performed to exam the consistency of the two coders' results before they discuss to address the disagreements.

2) *Impact on sponsees*: We study the impact of sponsorships on the *sponsees* from three aspects as follows.

a) *Impact on general contributions*: For the impact of attaining sponsorships on *sponsees*' contributions, we first proposed the following null-hypothesis:

H1.1.0: There is no significant difference in sponsees' public contributions before and after the developer being sponsored.

The test for *H1.1.0* involves *CMA* to *ISCD*, and the independent variable for this test is whether the developer is being sponsored. Since the metrics used in this test do not follow a normal distribution, we use a Wilcoxon Test which is suit for non-parametric repeated experiments with significance level $\alpha = 0.05$ to perform this hypothesis testing.

b) *Impact on the focus of sponsees' commits*: After the above preliminary analysis, we count the repositories where *sponsees* contributed commits to study whether sponsorships changes the focus of *sponsees*' contribution activities. We find 864 *sponsees* who had commits in the time window, and we

calculate the ratio of *sponsees* that committed in different developers' repositories. Next we count the months *sponsees* committed more in *sponsors*' repositories than themselves. If the number of months is more than 15, we think the *sponsee* commits most in his/her *sponsors*' repositories; otherwise, he/she commits most in selves repositories. And to get a statistical validation for this inference, we use a single-sided Wilcoxon test to conduct the following hypothesis testing:

H1.2.0: There is no significant difference in the number of commits authored by sponsees in different types of users' projects before and after being sponsored.

c) *Impact of the number of sponsors*: Existed studies [32] find that the level of reward also has an impact on the intensity of strength of individuals' responses (contributions level in this work). Since the amount of monthly income through GitHub Sponsors is private, we use the number of *sponsors* each month to represent the level of reward. To minimise the bias of individual contribution differences, we first conduct the normalization on the number of *sponsees*' monthly contribution activities. Then we apply the linear regression model on the normalized monthly contributions of *sponsees*, which estimates the effect of the number of *sponsors* with a treated variable and coefficient. The corresponding model is

$$y = \beta_0 + \beta_1 x \quad (1)$$

where y denotes *CMA* to *PRM* of a *sponsee* with the number of *sponsors* x in a month. We do not choose *ISCD* because there is no linear correlation between *ISCD* and the number of monthly *sponsors* intuitively.

3) *Impact on sponsors*: We study the influence of sponsorships on the *sponsors* with the following null-hypothesis.

H1.3.0: There is no significant difference in sponsors' public contributions before and after they provide sponsorships.

This hypothesis concerns the impact of sponsorship on *sponsors*' authored commits and issues because we want to focus on the code contributions change of *sponsors* to study whether there are more workload of them and on the questions (issues) raised by them to study if they want to 'hire' developers to solve their problems and feature requests.

In addition to *CMA* and *ISA* in Table II, we compute the number of commits and issues submitted to *sponsees*' projects as CMA_{Pse} and ISA_{Pse} , respectively. The *PRC* to *ISCD* are not included because the efficiency of *sponsors* reviewing issues/PRs is not important in this question. Then we calculate the ratio of *sponsors*' commits and issues submitted to *sponsees*' projects in their total commits and issues as *RCMA* and *RISA*. The formulations are as follows:

$$RCMA = \frac{CMA_{Pse}}{CMA}, \quad RISA = \frac{ISA_{Pse}}{ISA} \quad (2)$$

Moreover, we select the issues authored by *sponsors* and closed by their *sponsees* and calculate the close delay (like *ISCD*) as $ISCD_{Pse}$ to study whether *sponsees* have priorities on issues of *sponsors*.

So in the above hypothesis test, we use the value of CMA and ISA in Table II and CMA_{Pse} to $ISCD_{Pse}$ as the dependent variables, and the independent variable is whether the developer is providing sponsorship. Also a Wilcoxon Test is used to conduct this hypothesis testing, and the results are shown in Table IV.

4) *Impact on collaborators*: As GitHub is a collaborative platform, it is likely that a developer could motivate his/her collaborators after he/she got sponsorships. So we conduct statistical tests on the following null-hypothesis:

H1.4.0: There is no significant difference in public contributions of sponsees' collaborators before and after they were sponsored.

First, we define two periods for developer D and their collaborators C :

- Before: from Feb 1st, 2018 to the time D started being sponsored.
- After: from the time D started being sponsored to Jan 31st, 2021.

Since we are more concerned about the impact of a developer on his/her all collaborators, we regard a developer's collaborators as a whole and compute the average value of contributions. And the dependent variable of this test is the same as *H1.1.0*, the independent variable is period t in {Before, After}. Then we use a Mann-Whitney U test with significance level $\alpha = 0.05$ to perform this hypothesis testing because there may not be equal number of collaborators before and after the sponsee got sponsorships.

B. Results

1) *Survey*: After reviewing the 871 sponsees' sponsor pages, a code book is devised to categorize the impact of sponsorship on developers' activities. There are five codes ranging from **C0** to **C4** which are explained as follows.

- C1** *The developers states that he/she will become more active after receiving sponsorships.* For example "I'll spending more time on developing and contributing to OSSs." "motivate me to continue making ongoing contributions" "help me focus more on my open source work, increasing their development speed and overall quality"
- C2** *Priority support for issues and pull-requests submitted by sponsors.* For example "Your issues and pull-request will be reviewed with priority!" "Prioritization of a feature that you need in my projects." "You'll be able to influence the direction of [project name] development." "your GitHub issues involving development work will prioritized above other users' requests."
- C3** *Consulting or direct contributions to sponsors' repositories.* For example "I will spent 1 hour per month on a project / topic of your choice." "Once a month you have the opportunity to have one one-hour video chat..." "I will work one day/month with your team on an open-source project of your choice." "I'll fix a bug in your code (public or not)"

TABLE III
RESULT OF COHEN'S KAPPA COEFFICIENT AND THE RATE OF EACH CODE.

Code	Kappa	Rate
C0	4.48e-01	0.57%
C1	8.58e-01	46.84%
C2	7.93e-01	26.29%
C3	7.69e-01	33.87%
C4	7.51e-01	14.70%

C4 *Access to private repositories or exclusive channels / resources.* For example "A Patron license to any of my copyleft-licensed projects." "[I will] give you access to all of my private builds, tools and other personal stuff..."

In addition to the above categories, some developers clearly states that they will not be affected by sponsorships or provide rewards to their sponsors. For example "[Project name] does not offer any rewards for sponsors." "sponsorships will not yield any form of contract or obligation for me to meet". We also include this category in our code book as **C0**.

There are also rewards like sending gifts like stickers or t-shirts to the sponsors. We omit these items because the developer does not clearly state how sponsorship will affect his/her contributing activities. Next, based on the code book, two researchers (i.e., the first and second author) independently assign codes to all the 871 developers according to their sponsor pages. The coding results are shown in Table III.

Through the survey we find that nearly half of the sponsees state that receiving sponsorships motivates their contributions in GitHub, and less than 1% developers think sponsorship is not helpful of their activities. Moreover, some developers provide priorities to attract sponsors. Specifically, 26.29% developers promise they will prioritize sponsors' issues and PRs. 33.87% developers provide private services and 14.70% developers give sponsors access to exclusive resources, thus helping sponsors not only in sponsees' own repositories.

2) *Impact on developers*: We analyze the impact of sponsorships on three types of developers in this section.

a) *Impact on sponsees*: As described in Table II, we use CMA to $ISCD$ to measure developers' public contributions. The distributions of these numerical features are shown in Figure 4, and the results of statistical tests are shown in Table IV. Most metrics of sponsee show a significant less before developers got sponsorships, suggesting that developers generally contribute more after they get sponsorships. The test on ISC (The number of issues closed by the developer) and PRM (The number of PRs merged by the developer) shows that developers close fewer issues and merge fewer PRs. And the test on $ISCD$ (the first close delay of issues closed by the developer) shows that developers take more time to respond to (close) issues.

It is likely that users would contribute more with the development of GitHub, so we use the average contributions of all GitHub users as baseline to make comparisons. The results are shown in Fig. 5 From Fig. 5(a) we can see that the average number of commits made by all GitHub users nearly have no changes, while sponsees have more commits after t_{16} (after

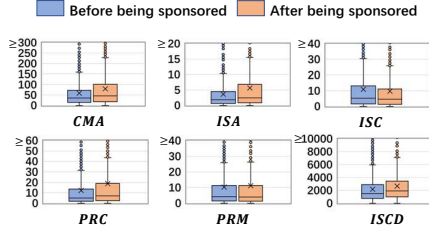


Fig. 4. Distributions of measures in Table II of developers before and after being sponsored.

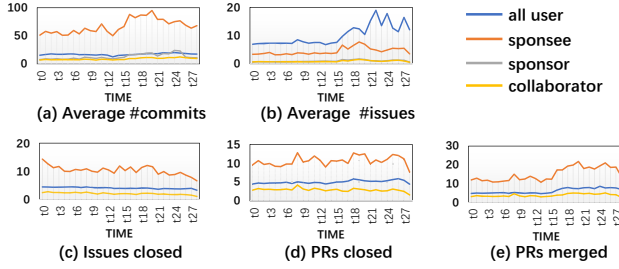


Fig. 5. The number of users' contributions each month from Feb. 2018 to Jan. 2021. (t_{15} and t_{16} corresponds to May. 2019 and Jan. 2020, respectively.)

TABLE IV

THE RESULTS FOR **RQ2**, PAIRED AND SINGLE-SIDED WILCOXON TEST AND MANN-WHITNEY TEST. SYMBOLS: '>' — SIGNIFICANTLY GREATER, '<' — SIGNIFICANTLY LESS, '=' — NO SIGNIFICANT DIFFERENCE. P-VALUES SHOWN NEXT TO THE SYMBOLS.

Subjects	Measure	Significance
Sponsee	<i>CMA</i>	<, 4.93e-16***
	<i>ISA</i>	<, 2.82e-18***
	<i>ISC</i>	>, 3.61e-09***
	<i>PRC</i>	<, 2.04e-16***
	<i>PRM</i>	>, 1.70e-03**
	<i>ISCD</i>	<, 7.71e-11**
Sponsor	<i>CMA</i>	<, 1.46e-08***
	<i>ISA</i>	<, 6.92e-26***
	<i>CMA_{Pse}</i>	=, 7.09e-01
	<i>ISA_{Pse}</i>	<, 7.34e-06***
	<i>RCMA</i>	=, 3.42e-01
	<i>RISA</i>	=, 1.04e-01
Collaborator	<i>ISCD_{Pse}</i>	<, 3.85e-06***
	<i>CMA</i>	=, 2.09e-01
	<i>ISA</i>	<, 1.78e-03**
	<i>ISC</i>	=, 1.13e-01
	<i>PRC</i>	<, 2.24e-02*
	<i>PRM</i>	=, 5.90e-01
	<i>ISCD</i>	<, 2.07e-25***

***p < 0.001, **p < 0.01, *p < 0.05

being sponsored). And in Fig. 5(b), the trend line of *sponsees* is similar to that of all GitHub users before t_{15} (before getting sponsorship). The average number of issues *sponsees* raised have been declining after t_{16} and is below the average of all GitHub users. Fig. 5(c-e) show that *sponsees* closed less issues and more PRs after t_{16} , and three contributions are all above the average of GitHub users. The average number of issues closed by all GitHub users nearly has no change with the passage of time, and the increase of PRs closed by them are

TABLE V

THE RESULTS FOR *H1.2.0*, PAIRED AND SINGLE-SIDED WILCOXON TEST USED (SIGNIFICANCE LEVEL $\alpha = 0.05$). SYMBOLS FOLLOW TABLE IV.

Measure	size	significance
<i>CMA_{Pself}</i>	699	<, 2.39e-02*
<i>CMA_{Psponsor}</i>	20	=, 5.46e-01
<i>CMA_{Pother}</i>	759	<, 9.75e-19***

TABLE VI

LINEAR REGRESSION MODEL ESTIMATES OF THE SPONSEES' MONTHLY CONTRIBUTIONS.

Measure	β_0	β_1	Coeff. of determination
<i>CMA</i>	86.4557	1.2439	0.03
<i>ISA</i>	5.2599	0.0560	0.00
<i>ISC</i>	6.9639	0.2738	0.11
<i>PRC</i>	14.7814	0.4106	0.02
<i>PRM</i>	9.1623	0.0201	0.03

less than that of *sponsees*.

As for the distribution of repositories where *sponsees* committed, we find that 85.3% of 864 *sponsees* have never committed in repositories owned by their *sponsors*, and 6.13% *sponsees* have never committed in their own repositories. And there are no *sponsee* committed only in their *sponsors*' repositories. In addition, among the 127 *sponsees* who have committed in both their own and their *sponsors*' repositories, we find that almost all (96%) *sponsees* have more commits in their own repositories in most months.

Then we use a Wilcoxon Test to show the differences in *sponsees*' commits in different developers' repositories, and the results are listed in Table V. We can see that *sponsees* have significantly more commits in their own repositories and others' repositories after being sponsored, while no significant difference in the number of commits in *sponsors*' repositories, suggesting that *sponsees* concentrate more on their own repositories after getting sponsorships.

The linear regression model in Sec. V-A2 estimates the *sponsees*' monthly contributions with the number of *sponsors*, and the results are shown in Table VI. The β_1 of 5 measures are all larger than 0, which means that the number of *sponsors* has a positive influence on all contribution activities. While the fourth column of this table shows that less than 10% of the y can be fitted by x , which suggests a weak positive impact.

b) *Impact on sponsors*: From Table IV we can see that most metrics show a significant less before developers provided sponsorships which means *sponsors* contribute more after they provided sponsorships. There is no significance difference in *CMA_{Pse}*, *RCMA* and *RISA* but a significance less in *CMA* and *ISA*, suggesting that *sponsors* become more active generally, not only in their *sponsees*' repositories. The *ISCD_{Pse}* shows significantly greater after developers provided sponsorships, meaning that *sponsees* responded to *sponsors*' issues more carefully.

c) *Impact on collaborators*: The results in Table IV show that sponsorships have no significant influence on *collaborators*' monthly commits (*CMA*), closed issues (*ISC*) and merged PRs (*PRM*), while the rest metrics of collaborators

TABLE VII
METRICS TO MEASURE A PROJECT'S OUTCOME AND QUALITY DURING A TIME WINDOW $\mathcal{L}$.

Category	Metric	Description
Outcome	PCM	The number of commits submitted to the project.
	PIS	The number of issues submitted to the project.
	$PISC$	The number of issues submitted to the project which were closed.
	$PPRC$	The number of PRs submitted to the project which were closed.
	$PPRM$	The number of PRs submitted to the project which were merged.
Quality	PIS_{bug}	The number of [bug report] issues submitted to the project.
	$PISC_{bug}$	The number of [bug report] issues submitted to the project which were closed.
	$PISC_{D_{bug}}$	The average time delay of first closing a [bug report] issue submitted to the project.

show the same changes with that of *sponsees*.

Summarizing the above results, we reject all the null-hypotheses raised in RQ2 and draw the conclusion: *sponsees have more contributions and take issues/PRs more seriously after being sponsored. And not only sponsees, sponsors and collaborators can also be positively influenced by sponsorships. Moreover, most sponsors provide sponsorships to support sponsees own work, instead of hiring sponsees to work for them. However, the increase in the number of sponsors are not accompanied with the increase in sponsees' contributions.*

VI. IMPACT ON THE OUTCOME AND QUALITY OF SPONSEES' PROJECTS

Open source projects are the cornerstone of the OSS, and the cooperation of developers is essential for the success of most projects on GitHub. In this section we answer **RQ3: What are the impacts of sponsorships on the outcomes and qualities of projects related to developers who got sponsorships?** From the result of **RQ2** we find that *sponsees* expend most of their contributions on their own work and contribute more after they were sponsored, which supports our assumption that sponsorships also motivate projects related to *sponsees*.

And in the following sections, we separate the projects owned by *sponsees* and those contributed by *sponsees* to conduct studies respectively.

A. Research Methods

According to [28], [33], [34] we choose two aspects of projects-*outcome* and *quality* to measure whether they are motivated. And the metrics we used are listed in Table VII.

For each *sponsee*, we count all projects outcomes/qualities of his/her. And for the projects contributed by *sponsees*, we select those where there are at least 1 commit every month in our time window (the number of the remained projects is 2318) to ensure they are active. To test the statistical significance of the change of the above measures, we raise the following hypotheses:

H2.1.0: There is no significant difference in sponsees' project outcomes before and after they were sponsored.

TABLE VIII
THE RESULTS FOR *H2.1.0* AND *H2.2.0*, PAIRED AND SINGLE-SIDED MANN-WHITNEY TEST. SYMBOLS FOLLOW TABLE IV.

Subjects	Measure	Significance
Outcome	PCM	$<, 2.66e-04^{***}$
	PIS	$<, 1.01e-13^{***}$
	$PISC$	$<, 5.47e-03^{**}$
	$PPRC$	$<, 9.62e-22^{***}$
	$PPRM$	$<, 4.08e-06^{***}$
Quality	PIS_{bug}	$<, 6.70e-03^{**}$
	$PISC_{bug}$	$<, 9.50e-03^{**}$
	$PISC_{D_{bug}}$	$<, 5.04e-07^{***}$
***p < 0.001, **p < 0.01, *p < 0.05		

H2.2.0: There is no significant difference in the quality of projects owned by sponsees before and after being sponsored.

H2.3.0: There is no significant difference in the outcome of projects where sponsees contributed before and after they were sponsored.

H2.4.0: There is no significant difference in the quality of projects where sponsees contributed before and after they were sponsored.

The dependent variable is the total value of metrics listed in Sec. VI of projects each month, and the independent variable for this test is whether the developer/project is being sponsored. For *H2.3.0* and *H2.4.0*, if more than 1 *sponsee* contributed in the same project, then the earliest sponsored time of them is used to determine whether the project is sponsored or not.

As the projects involved in *H2.1.0* and *H2.2.0* in two groups are not the same, so we use a Mann-Whitney U test with significance level $\alpha = 0.05$ to perform these hypothesis testing. In *H2.3.0* and *H2.4.0*, the size of two group are equal. While the metrics used do not follow a normal distribution, we use a Wilcoxon Test which is suit for nonparametric repeated experiments with significance level $\alpha = 0.05$ to perform these hypotheses testing.

B. Results

a) *Impact on the projects owned by sponsees:* The results in Table VIII show that there is significantly less in all metrics of *outcome* before the project's owner got sponsorships. After a developer became a *sponsee*, issues raised and closed, and PRs closed and merged in his/her projects were all more than before. And for the measures of *quality*, we can see that there is a significantly less in three metrics before and after the project's owner was sponsored. Thus from this change we can infer that developers in *sponsees'* projects are more active and review issues and PRs more carefully.

b) *Impact on the projects where sponsees have ever contributed:* From the results shown in Table IX we can see that the PRM of projects after their contributors got sponsored are significantly less than before, while the rest 7 metrics show a contrary significance. That is, there are more outcome and higher quality of projects after their contributors got sponsorships. So we can infer that sponsorships giving

TABLE IX
THE RESULTS FOR H2.3.0 AND H2.4.0, PAIRED AND SINGLE-SIDED
WILCOXON TEST. SYMBOLS FOLLOW TABLE IV.

Subjects	Measure	Significance
Outcome	PCM	<, 3.06e-05***
	PIS	<, 4.22e-47***
	PISC	<, 3.31e-10***
	PPRC	<, 2.25e-112***
	PPRM	>, 4.83e-03**
Quality	PIS _{bug}	<, 7.32e-07***
	PISC _{bug}	<, 6.43e-03**
	PISC _{D_{bug}}	<, 4.79e-90***
***p < 0.001, **p < 0.01, *p < 0.05		

to developers also have a significant incentive on their contributed projects.

We reject the above null-hypotheses and draw the conclusion: *sponsorships not only play a positive part in motivating developers' own projects, but also can promote the outcome and quality of projects where sponsees have ever contributed.*

VII. DISCUSSION

In this section, we discuss the implications and the threats to validity of our work.

A. Implications

In our work, we present a data set of monthly sponsorship snapshots and activities of related users and projects which can be used in future works.

In Sec. V-A1, we proposed a context analysis method to do a objective survey of *sponsees* as possible. And we summarize the attitude of developers towards GitHub Sponsors program, which is helpful for the platform managers to improve the financial support mechanism of GitHub.

In Sec. V, we find that *sponsees* are significantly more active on GitHub after being sponsored, and they are more concentrated on their own work especially. While the number of *sponsors* seems to have little impact on the contributions of *sponsees*. In addition, providing sponsorships also motivates the *sponsors* and *collaborators*. The results can inspire future researchers to compare contributors who are sponsored with those who are not sponsored, thus helping OSS community to better manage their contributors and improve the sustainability of the community. Moreover, the outcome and quality of projects related to *sponsees* are significantly improved after they got sponsorships. Although sponsoring projects on GitHub can only use external channels now, it is interesting to study whether sponsoring projects or sponsoring projects' contributors motivates the outcome and quality more if data available.

B. Threats to validity

Internal Validity. In this work, we conduct a data-driven approach to find the impact of sponsorships on developers' contributions, responses and their related projects' outcomes and qualities. However, we can not eliminate the potential effects of other factors, such as other sources of financial supports [15], [19], [21], users' roles in the projects [35],

user collaboration and interactions [36], [37], etc to the users' activities. As a result, we point out that the conclusions made in this work do not necessarily imply a causal relationship between sponsorships and the user' activity changes.

External Validity. Because the GitHub Sponsors program is relatively new to the community, currently there are a limited number of participants that can be involved in our study. As a result, the conclusions made in this paper may not generalize well beyond the current scope and context of the data set. However, GitHub Sponsors is developing and it is hopeful that we can improve the generalization ability of the results in this paper, and develop robust theories, with more data.

Construct Validity. In this work, we operationally define the concept of users' *response* with the delay of first closing a issue as presented in Sec. V. The operational definition is not guaranteed to be good or complete to characterize the users' responses. And in Sec. V, we use the number of *sponsors* to represent the level of reward, while it is not as suitable as the precise amount of the income. So it may bring bias on the result of our linear regression model. While we believe that it is straight-forward to extend the features and measures used in this paper using more data from the software repositories.

VIII. CONCLUSION

In this paper, we present a data-driven research on the newly emerged GitHub Sponsors program which aims at bringing financial supports to the OSS developers. We make a preliminary profile of users taking part in this program, and find the difference and similarities between *sponsees* and *sponsors*. With the guidance of the overjustification theory, we analyze and the changes of users' contributions and responses over time and discover that they become significantly more active after getting sponsorships. And we also find that providing sponsorships to developers can improve their own or contributed projects. Although there are many factors leading that the results of users' responses are not sound inspiring, we can improve our method and conduct more experiments when there is enough data we collect. The findings in this work are potentially useful for motivating future researches, and helping to improve the operation of the programs.

As future work, we plan to compare sponsored developers with similar developers without sponsorships. And we will considerate the other sources of financial income like Open-collective and Patreon. We also intend to study the impact of other stimuli like reputation on developers' activities, not limited to monetary motivation.

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